Full Mock Test 2 · Paper 1

Question 1

What is the value of

$$\left(\left((2+1)^{-1}+1\right)^{-1}+1\right)^{-1}+1$$

A $\frac{5}{8}$

B $\frac{11}{7}$

C $\frac{8}{5}$

D $\frac{18}{11}$

E $\frac{15}{8}$

Worked solution · B

Work outwards. The successive quantities after adding 1 are 4/3, 7/4 and 11/7.

Full Mock Test 2 · Paper 1

Question 2

For how many (not necessarily positive) integer values of n is the value of $4000 \cdot \left(\frac{2}{5}\right)^n$ an integer?

A 4
B 5
C 6
D 7
E 8
F 9
G 10
H 11

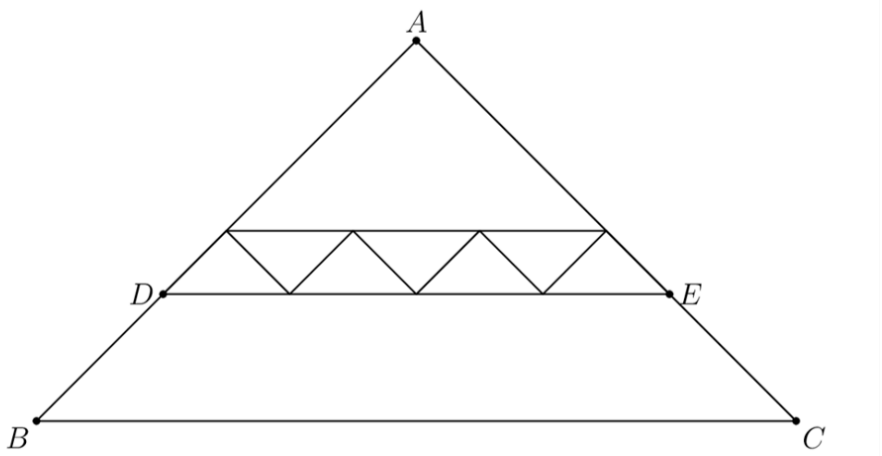
Worked solution · F

Write the expression as $2^{5+n} \times 5^{3-n}$. Both exponents must be non-negative. The integers from -5 to 3 inclusive give nine values.

Full Mock Test 2 · Paper 1

Question 3

All of the triangles in the diagram below are similar to isosceles triangle $\triangle ABC$, in which $AB = AC$. Each of the 7 smallest triangles has area 1, and $\triangle ABC$ has area 40. What is the area of trapezoid $DBCE$?



A 16

B 18

C 20

D 22

E 24

Worked solution · E

The base DE is four small-triangle bases wide. By similarity, triangle ADE has sixteen times the area of a smallest triangle, so the trapezoid has area 40 − 16 = 24.

Full Mock Test 2 · Paper 1

Question 4

Suppose that real number x satisfies

$$\sqrt{49 - x^2} - \sqrt{25 - x^2} = 3$$

What is the value of $\sqrt{49 - x^2} + \sqrt{25 - x^2}$?

A 8

B $\sqrt{33} + 8$

C 9

D $2\sqrt{10} + 4$

E 12

Worked solution · A

Multiply the given difference by the unknown sum. The product is (49 − x²) − (25 − x²) = 24, so the sum is 8.

Full Mock Test 2 · Paper 1

Question 5

Two open intervals (−1, 3) and (k − 1, k + 1) overlap in an interval of length exactly 1. What is the sum of the squares of all possible real values of k?

A 10

B 4

C 5

D 8

E 2

Worked solution · A

If the shorter interval sticks out to the left, its right endpoint must be 0, so k = −1. If it sticks out to the right, its left endpoint must be 2, so k = 3. If it is contained inside, the overlap has length 2. The required sum is 1 + 9 = 10.

Full Mock Test 2 · Paper 1

Question 6

Two upright cylindrical vessels have internal radii 2r and 3r. Initially the water in each is h deep. Water is poured from the narrower vessel into the wider one until the depth in the wider vessel is twice that in the narrower one. What fraction of the narrower vessel's original water is poured out?

A 1/3

B 4/11

C 9/22

D 1/2

E 9/13

Worked solution · C

Let f be the fraction transferred. The narrower vessel loses depth fh; the wider one gains depth 4fh/9, because their base areas are in the ratio 4:9. Thus 1 + 4f/9 = 2(1 − f), giving f = 9/22.

Full Mock Test 2 · Paper 1

Question 7

A convergent geometric progression has a positive first term and a negative second term. Its sum from the third term onwards is one quarter of its total sum. Its third and fourth terms add to 3. What is the total sum?

A 8

B 12

C 24

D 18

E 16

Worked solution · E

The tail starting at the third term is the entire series multiplied by r^2, so $r^2 = 1/4$. The sign condition gives $r = -1/2$. If a is the first term, the third and fourth add to $a(1/4 - 1/8) = 3$, so $a = 24$. The total is $24/(1 + 1/2) = 16$.

Full Mock Test 2 · Paper 1

Question 8

How many ordered pairs of real numbers (x, y) satisfy the following system of equations?

$$x + 3y = 3$$

$$\big|\,|x| - |y|\,\big| = 1$$

A 1
B 2
C 3
D 4
E 5
F 6
G 7
H 8

Worked solution · C

The equality requires either x − y = ±1 or x + y = ±1. Intersecting these four lines with x + 3y = 3 gives (3/2, 1/2), (0, 1) twice, and (−3, 2). All three distinct points satisfy the equation.

Full Mock Test 2 · Paper 1

Question 9

The coefficient of x^2 in a quadratic $f(x)$ is 1, and its two zeros are 2 units apart. Its graph meets the reflected graph $y = f(4 - x)$ at $(2, 8)$. What is the sum of all possible values of $f(0)$?

A 12

B 16

C 24

D 20

E 32

Worked solution · C

Write $f(x) = (x - h)^2 - 1$: the zeros are $h - 1$ and $h + 1$. On the mirror line $x = 2$ the graphs agree, so $(2 - h)^2 = 9$. Thus h is -1 or 5, giving $f(0)$ equal to 0 or 24.

Full Mock Test 2 · Paper 1

Question 10

A circle passes through (0, 0), (8, 0) and B = (0, b), where b > 0. The tangent at B passes through (3, 10). Two values of b are possible. What is their difference?

A 1

B 4

C 3

D 2

E 6

Worked solution · D

The centre is (4, b/2), from the perpendicular bisectors of the two chords on the axes. The gradient of the radius to B is −b/8, so the tangent has gradient 8/b. Equating this to (10 − b)/3 gives $b^2 - 10b + 24 = 0$. The two heights are 4 and 6.

Full Mock Test 2 · Paper 1

Question 11

A real number a is chosen so that the coefficients of x^3 and x^5 in $(1 + x)^6(1 - ax^2)$ are equal. What is the coefficient of x^4 in that product?

A 0

B 30

C 15

D 28

E 14

Worked solution · B

The two coefficients are $20 - 6a$ and $6 - 20a$, respectively, so $a = -1$. With that value the x^4 coefficient is the sum of the x^4 and x^2 coefficients of $(1 + x)^6$, namely $15 + 15$.

Full Mock Test 2 · Paper 1

Question 12

A point at angle x moves once round the unit circle, with $0 \le x < 2\pi$. For what fraction of the full turn is the point above the horizontal axis while $\cos(2x) > \cos x$?

A 1/12

B 1/4

C 1/6

D 1/3

E 1/2

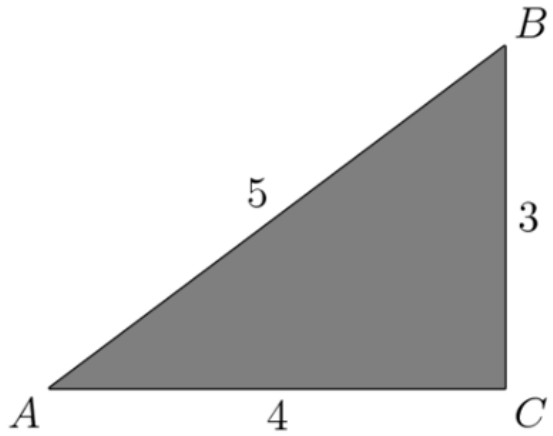
Worked solution · C

Above the axis means $0 < x < \pi$. Writing $c = \cos x$, the inequality is $(2c + 1)(c - 1) > 0$. Since $c < 1$ in this interval, it requires $c < -1/2$, or $2\pi/3 < x < \pi$. This arc has length $\pi/3$ out of 2π.

Full Mock Test 2 · Paper 1

Question 13

A paper triangle with sides of lengths 3, 4, and 5 inches, as shown, is folded so that point A falls on point B. What is the length in inches of the crease?



A $1+\frac{1}{2}\sqrt{2}$

B $\sqrt{3}$

C $\frac{7}{4}$

D $\frac{15}{8}$

E 2

Worked solution · D

Put A = (0, 0), B = (4, 3) and C = (4, 0). The crease lies on the perpendicular bisector of AB, from (2, 3/2) to (25/8, 0). Its length is the hypotenuse with legs 9/8 and 3/2, giving 15/8.

Full Mock Test 2 · Paper 1

Question 14

Real numbers a and b satisfy $2^a + 2^b = 12$ and $2^{(a+b)} = 32$. What is the value of $2^{(a-b)} + 2^{(b-a)}$?

A 3/2

B 5/2

C 9/4

D 2

E 4

Worked solution · B

Set $u = 2^a$ and $v = 2^b$, so $u + v = 12$ and $uv = 32$. The requested sum is $(u^2 + v^2)/(uv) = (144 - 64)/32 = 5/2$. The data are attainable: u and v are 4 and 8.

Full Mock Test 2 · Paper 1

Question 15

A rectangular plot has sides parallel to the coordinate axes. Its upper-right corner is (6, 5). Its lower-left corner (u, v) is in the first quadrant on the line u + 2v = 8. What is the greatest possible area of the plot?

A 8

B 7

C 6

D 9

E 10

Worked solution · A

Substitute v = 4 − u/2. The area is (6 − u)(5 − v) = (6 − u)(1 + u/2) = 8 − (u − 2)²/2. This is at most 8. The corner (2, 3) lies on the required line and attains the bound.

Full Mock Test 2 · Paper 1

Question 16

For a fixed real number c, both inequalities $2^{x-c} \le M$ and $4^{c-x} \le M$ must hold for every x in [−1, 5]. What is the smallest M obtainable by choosing c?

A 4

B 8

C 32

D 16

E 64

Worked solution · D

The first expression is largest at x = 5, the second at x = −1. Thus M must be at least the larger of 2^{5-c} and 2^{2c+2}. The first decreases in c and the second increases, so their maximum is smallest at equality. This gives c = 1 and M = 16.

Full Mock Test 2 · Paper 1

Question 17

Which of the following describes the set of values of a for which the curves $x^2 + y^2 = a^2$ and $y = x^2 - a$ in the real xy-plane intersect at exactly 3 points?

A $a = \frac{1}{4}$

B $\frac{1}{4} < a < \frac{1}{2}$

C $a > \frac{1}{4}$

D $a = \frac{1}{2}$

E $a > \frac{1}{2}$

Worked solution · E

Substitution gives x²(x² + 1 − 2a) = 0. There are three distinct x-values exactly when 2a − 1 > 0. Each gives just one y, so a > 1/2.

Full Mock Test 2 · Paper 1

Question 18

A function f is defined recursively by

$$f(1) = f(2) = 1 \quad \text{and} \quad f(n) = f(n-1) - f(n-2) + n$$

for all integers $n \geq 3$. What is $f(2018)$?

A 2016

B 2017

C 2018

D 2019

E 2020

Worked solution · B

Set g(n) = f(n) − n − 1. Then g(n) = g(n − 1) − g(n − 2), starting with −1, −2. Its repeating block is −1, −2, −1, 1, 2, 1, as the first pair then recurs. Since 2018 has remainder 2 on division by 6, f(2018) = 2018 + 1 − 2 = 2017.

Full Mock Test 2 · Paper 1

Question 19

In the xy-plane, a parabola has vertex $(9, -14)$ and intersects the x-axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a, b, and c are constants, which of the following could be the value of $a + b + c$?

A -23

B -19

C -14

D -12

Worked solution · D

The parabola is y = a(x − 9)² − 14 with a > 0, since it has two zeros. Thus a + b + c = f(1) = 64a − 14 can take any value greater than −14. Of the options only −12 qualifies.

Full Mock Test 2 · Paper 1

Question 20

What is

$$\sum_{i=1}^{100}\sum_{j=1}^{100}(i+j)?$$

A $100{,}100$

B $500{,}500$

C $505{,}000$

D $1{,}001{,}000$

E $1{,}010{,}000$

Worked solution · E

Each value from 1 to 100 occurs 100 times as i and another 100 times as j. The sum is 200 × (100 × 101/2) = 1,010,000.

Full Mock Test 2 · Paper 2

Question 1

The sum of two nonzero real numbers is 4 times their product. What is the sum of the reciprocals of the two numbers?

A 1
B 2
C 4
D 8
E 12

Worked solution · C

If the two numbers are u and v, then u + v = 4uv. Divide by the nonzero product uv to get 1/u + 1/v = 4.

Full Mock Test 2 · Paper 2

Question 2

Alice, Bob, and Charlie were on a hike and were wondering how far away the nearest town was. When Alice said, "We are at least 6 miles away," Bob replied, "We are at most 5 miles away." Charlie then remarked, "Actually the nearest town is at most 4 miles away." It turned out that none of the three statements were true. Let d be the distance in miles to the nearest town. Which of the following intervals is the set of all possible values of d?

A $(0, 4)$
B $(4, 5)$
C $(4, 6)$
D $(5, 6)$
E $(5, \infty)$

Worked solution · D

Alice being wrong gives d < 6. Bob being wrong gives d > 5, which already makes Charlie wrong. Hence 5 < d < 6.

Full Mock Test 2 · Paper 2

Question 3

Ms. Carroll promised that anyone who got all the multiple choice questions right on the upcoming exam would receive an A on the exam. Which one of these statements necessarily follows logically?

A If Lewis did not receive an A, then he got all of the multiple choice questions wrong.

B If Lewis did not receive an A, then he got at least one of the multiple choice questions wrong.

C If Lewis got at least one of the multiple choice questions wrong, then he did not receive an A.

D If Lewis received an A, then he got all of the multiple choice questions right.

E If Lewis received an A, then he got at least one of the multiple choice questions right.

Worked solution · B

Take the contrapositive of the promise. Anyone who did not receive an A must have missed at least one multiple-choice question.

Full Mock Test 2 · Paper 2

Question 4

At a gathering of 30 people, there are 20 people who all know each other and 10 people who know no one. People who know each other hug, and people who do not know each other shake hands. How many handshakes occur?

A 240

B 245

C 290

D 480

E 490

Worked solution · B

Each of the ten people shakes hands with all twenty in the other group. Within the group of ten there are 10 × 9/2 more handshakes. The total is 200 + 45 = 245.

Full Mock Test 2 · Paper 2

Question 5

For an integer $n \geq 4$, a student compares the coefficient of x^4 in $(1 + x)^n$ with twice the coefficient of x^2 in $(1 + x)^{n-1}$. What is the least n for which the first quantity is strictly larger?

A 5

B 6

C 8

D 7

E 9

Worked solution · D

Dividing the coefficient of x^4 by the unscaled coefficient of x^2 gives $n(n - 3)/12$, by cancellation of factorials. The strict inequality therefore requires $n(n - 3) > 24$. The left side increases for $n \geq 4$; it is 18 at $n = 6$ and 28 at $n = 7$.

Full Mock Test 2 · Paper 2

Question 6

A student claims: ‘If 6 divides n(n + 1), then 6 divides at least one of n and n + 1.’ How many integers n from 1 to 60 inclusive are counterexamples?

A 10

B 15

C 50

D 30

E 40

F 45

G 20

Worked solution · G

The consecutive factors supply a factor 2 automatically. For a counterexample a factor must supply 3 without either factor supplying 6. This happens exactly when n is 2 or 3 modulo 6 (the odd multiple of 3 is n + 1 or n). Each residue occurs ten times.

Full Mock Test 2 · Paper 2

Question 7

A trapezium-rule estimate for the integral of a quadratic $ax^2 + bx + c$, with $a > 0$, uses just two strips: [0, 1] and [1, 3]. Halving only the first strip reduces the estimate by D. By how much would halving only the second strip reduce the original estimate?

A 2D

B 8D

C 4D

D 6D

E 3D

Worked solution · B

For a strip of width h starting at t, subtracting the two half-strip estimates from the original gives $h[f(t) − 2f(t + h/2) + f(t + h)]/4$. The bracket is $ah^2/2$. Thus the reduction is $ah^3/8$, independent of t, and doubling the width multiplies it by 8.

Full Mock Test 2 · Paper 2

Question 8

Real numbers u < v are fixed. For every straight-line function f(x) = mx + c, the integral of f from u to v equals (v − u)f(1). Which condition is necessary and sufficient for this property?

A u + v = 2

B u ≤ 1 ≤ v

C uv = 1

D v − u = 2

E u = 0 and v = 2

Worked solution · A

The mean value of a linear function on [u,v] is m(u + v)/2 + c. For this to equal m + c for every slope requires (u + v)/2 = 1 (take m = 1). That same condition works for every m and c.

Full Mock Test 2 · Paper 2

Question 9

P(n) is a statement about positive integers. You know P(4) is true, and whenever P(n) is true, both P(n + 4) and P(n + 6) are true. What is the greatest even integer at least 4 for which these facts do not force P(n) to be true?

A 12

B 8

C 10

D 6

E There is no greatest such integer

Worked solution · D

The rules give 8 and 10. Repeated additions of 4 from those two starting points give every even integer at least 8. They do not force 6: the set consisting of 4 and all even integers at least 8 obeys both rules but omits 6.

Full Mock Test 2 · Paper 2

Question 10

For $x > 1$ put $a = \log_2 x$ and $b = \log_x 8$. Which value of k makes the equation $a + b = k$ have exactly one possible value of x?

A $\sqrt{3}$

B 3

C $2\sqrt{3}$

D 4

E 6

Worked solution · C

We have $a > 0$ and $ab = 3$, so $a + b = a + 3/a \geq 2\sqrt{3}$. Equality requires $a = \sqrt{3}$ and determines x uniquely. Above this minimum the quadratic $a^2 - ka + 3 = 0$ has two positive roots; below it there are none.

Full Mock Test 2 · Paper 2

Question 11

An integer d > 0 divides n(n + 1)(2n + 1) for every positive integer n. For how many different values of d is this possible?

A 2

B 3

C 6

D 4

E Infinitely many

Worked solution · D

Taking n = 1 gives 6, so d must divide 6. The product is always even. For n congruent to 0, 1 or 2 modulo 3, respectively, n, 2n + 1 or n + 1 supplies a factor 3. Hence every divisor of 6 works: 1, 2, 3 and 6.

Full Mock Test 2 · Paper 2

Question 12

A function f permutes {1, 2, 3, 4}. For every x in this set, f(f(x)) = 5 − x. It is also known that f(1) = 2. What is f(3)?

A 4

B 2

C 3

D 1

E No such permutation exists

Worked solution · D

From x = 1 we obtain f(2) = 4; from x = 2, f(4) = 3; and from x = 4, f(3) = 1. This closes a four-cycle. The remaining condition at x = 3 is satisfied since f(f(3)) = f(1) = 2.

Full Mock Test 2 · Paper 2

Question 13

Each card shows a number and may carry a star, a square, both or neither. Every card with both symbols shows an even number. At least one card shows an odd number and carries a star. Which conclusion follows?

A Every card with a star lacks a square.

B Some odd-numbered card carries a square.

C Some even-numbered card carries both symbols.

D Every odd-numbered card carries a star.

E Some card with a square lacks a star.

F Every even-numbered card carries a square.

G Some card with a star lacks a square.

Worked solution · G

Choose an odd-numbered starred card whose existence is given. It cannot carry a square, because two symbols would force its number to be even. The premises do not require any squares or even cards. They also allow another card with both symbols and an even number, or an odd card with no star.

Full Mock Test 2 · Paper 2

Question 14

For $0 \le a \le 2$ let $I(a)$ be the integral of $|x - a|$ from 0 to 2. Which condition is equivalent to $I(a) \ge 5/4$?

A $|a - 1| \le 1/4$

B $|a - 1| \le 1/2$

C $|a - 1| \ge 1/4$

D $|a - 1| \ge 1/2$

E $|a - 1| \ge 3/4$

Worked solution · D

Split the area at $x = a$. The two triangles have total area $[a^2 + (2 - a)^2]/2 = 1 + (a - 1)^2$. Thus $I(a) \ge 5/4$ exactly when $(a - 1)^2 \ge 1/4$, with both boundary values included.

Full Mock Test 2 · Paper 2

Question 15

For a positive integer n, 72 divides n^2. What is the smallest possible value of the greatest common divisor of n and 72?

A 6

B 8

C 24

D 18

E 12

Worked solution · E

Since $72 = 2^3 \times 3^2$, n must have at least two factors of 2 and one of 3. Thus 12 divides both n and 72, so their greatest common divisor is at least 12. Taking n = 12 attains the bound because 72 divides 144.

Full Mock Test 2 · Paper 2

Question 16

For real x and y, consider the pair of conditions $|x + y| < 1$ and $|x − y| < 1$. Which single condition is equivalent to the pair?

A $|x| + |y| < 1$

B $x^2 + y^2 < 1$

C $|x| < 1$ and $|y| < 1$

D $|xy| < 1/4$

E $||x| − |y|| < 1$

Worked solution · A

When x and y have the same sign, $|x + y| = |x| + |y|$; when they have opposite signs, $|x − y|$ equals that sum. The other expression is never larger. Hence the maximum of the two is $|x| + |y|$, including when either variable is zero.

Full Mock Test 2 · Paper 2

Question 17

Joey and Chloe and their daughter Zoe all have the same birthday. Joey is 1 year older than Chloe, and Zoe is exactly 1 year old today. Today is the first of the 9 birthdays on which Chloe's age will be an integral multiple of Zoe's age. What will be the sum of the two digits of Joey's age the next time his age is a multiple of Zoe's age?

A 7

B 8

C 9

D 10

E 11

Worked solution · E

Let t be Chloe’s age minus Zoe’s age. Chloe’s age is a multiple of Zoe’s exactly when Zoe’s age divides t, so t has nine positive divisors. The stated two-digit age for Joey implies $t < 100$. The only such t is 36: a number with nine divisors is p^8 or p^2q^2 for distinct primes, and only $(2 \times 3)^2$ is below 100. Joey’s age exceeds Zoe’s by 37. After today the next eligible age for Zoe is 37, so Joey will be 74 and the digit sum is 11.

Full Mock Test 2 · Paper 2

Question 18

A token starts at 0. Each move adds either 2 or 5, and two moves of 5 may not be consecutive. In how many different ordered sequences of moves can the token finish at exactly 19?

A 5

B 7

C 9

D 8

E 10

Worked solution · C

With k moves of 5, 19 − 5k must be non-negative and even, so k is 1 or 3. For k = 1 there are seven moves of 2 and eight positions for the 5. For k = 3 there are two moves of 2; separating the fives forces the single order 5, 2, 5, 2, 5. Total: 9.

Full Mock Test 2 · Paper 2

Question 19

Let x range over [−2, 4] and m over {−1, 2, 4}. Let c be the least radius such that for every x there exists an m with |x − m| ≤ c. Let d be the least radius such that there exists one m with |x − m| ≤ d for every x. What is (c, d)?

A (1, 3)

B (3/2, 3)

C (2, 3)

D (3/2, 4)

E (2, 4)

Worked solution · D

With m allowed to vary, the largest interior gap between centres is 3. Its midpoint, x = 1/2, needs radius 3/2. This covers both gaps and the left endpoint, which is 1 from −1. With m fixed, the endpoint distances give worst cases 5, 4 and 6 for m = −1, 2 and 4 respectively. The smallest is 4.

Full Mock Test 2 · Paper 2

Question 20

For which values of k does there exist a quadratic f such that f(0) = f(2) = k, f′(x) > 0 throughout 0 < x < 1, and the integral of f from 0 to 2 is zero?

A k = 0

B k ≤ 0

C k < 0

D k > 0

E All real k

Worked solution · C

Write f(x) = k + ax(x − 2), with a nonzero. Its derivative is 2a(x − 1), so the required sign forces a < 0. Its integral is 2k − 4a/3; setting this to zero gives k = 2a/3 < 0. Conversely, for every negative k, choosing a = 3k/2 gives all three properties.